Experimental Design Types in Stated Choice Studies

By : ngene User Manual & Reference Guide

Presented by

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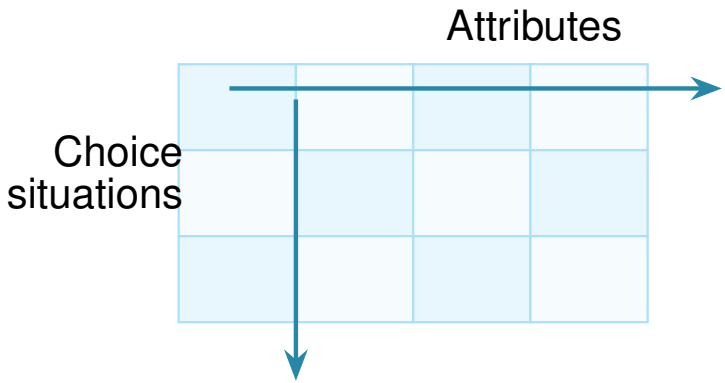
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Outlines

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2. Why Experimental Design Matters ?
3. Full Factorial Design
4. Orthogonal Design
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5. Worked Examples
6. Orthogonal blocking statistics of UBER project report
   - General statistics
   - Distribution From SPSS

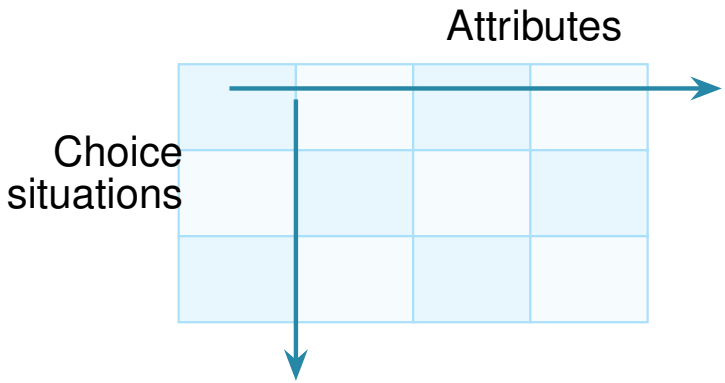
Core Idea

An experimental design may be viewed as a **matrix of values** that determines what goes where in a stated choice (SC) survey. Its rows and columns represent the **choice situations**, **attributes**, and **alternatives** of the experiment.



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An experimental design may be viewed as a **matrix of values** that determines what goes where in a stated choice (SC) survey. Its rows and columns represent the **choice situations**, **attributes**, and **alternatives** of the experiment.



Layout of the design matrix can be set out in two ways:

Option 1

Rows → choice situations
Columns → attributes

Option 2

Columns → choice situations
Rows → attributes

Motivating Example: Option 1

Rows → choice situations **Columns → attributes**

	car		train	
	Time	Cost	Time	Cost
1	20	5	30	4
2	25	6	25	5
3	25	5	30	3
4	...	...	...	...

Each **row** is a complete choice situation: a respondent sees the full attribute profile for *car* and *train* side by side and picks one. This is the natural "wide format" layout — one row per choice task, all attribute levels for all alternatives read off horizontally.

Motivating Example: Option 2

Columns → choice situations Rows → attributes

Choice situation 1

	car	train
Travel time (mins)	20	30
Fuel costs / fare ($)	5	4

Choice situation 2

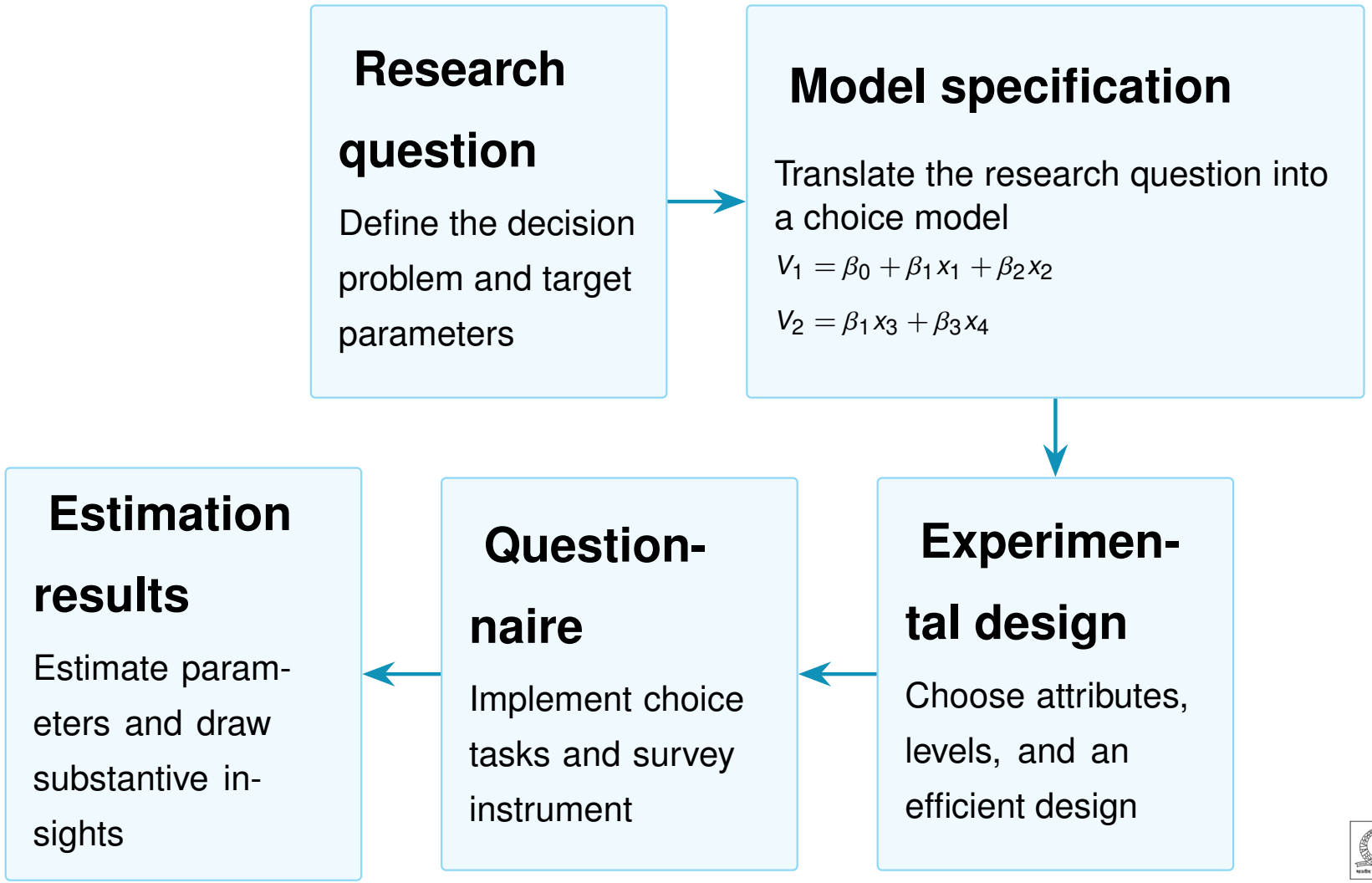
	car	train
Travel time (mins)	25	25
Fuel costs / fare ($)	6	5

Choice situation 3

	car	train
Travel time (mins)	25	30
Fuel costs / fare ($)	5	3

Here each choice situation is its own small **block**: **rows** hold the attributes (travel time, fare) and **columns** hold the alternatives. Stacking these blocks side by side effectively makes choice situations the ”column” dimension of the overall design.

From Question to Estimation



What Good Design Should Achieve



Coverage of relevant trade-offs

Include the attributes and levels needed to answer the research question.



Attribute-level balance

Ensure a balanced and statistically efficient assignment of attribute levels across choice tasks.



Low correlation among columns

Minimize multicollinearity so effects can be separately identified.



Manageable respondent burden

Keep tasks realistic in length and complexity to maintain data quality.

Main Design Families

1 Full factorial design

Includes every combination of attribute levels

2 Fractional factorial design

Uses a subset of combinations

3 Orthogonal design

Balances levels and removes correlation

4 Efficient design

Minimizes standard errors using priors

5 Orthogonal-in-differences (OOD)

Orthogonality in differences across alternatives

In modern applied work, efficient designs are often preferred when prior information is available.

Different Design Details

Design Type	Main Requirements	Pros	Cons
Full factorial	Very small attribute space	i Complete information ii All possible effects	i Grows too large fast ii Heavy respondent burden
Fractional factorial	Clear choice of subsets & important effects	i Practical size ii Flexible	i Maybe week if poorly selected ii Possible confounding
Orthogonal	Attribute level balance & uncorrelated columns	i Simple & transparent ii Independent estimation of main effects	i May create unrealistic tasks ii Orthogonality can be lost in real data
Efficient design	Prior parameter values & software based optimization	i Lower standard error ii Better statistical efficiency	i Depends on prior quality ii More complex to explain & generate
OOD	Common attributes across unlabeled alternatives	i Strong trade offs ii Useful specialized case	i Limited scope ii Can produce strong contrasts

Computation

Proposition & Formulation

If there are, J alternatives, each with K_j attributes and each attribute has l_{jk} levels, $\forall$ k $\in$ K_j. Let S^{ff} be total nos of full factorial scenario -

$$S^{ff} = \prod_{j=1}^{J} \prod_{k=1}^{k_j} l_{jk}$$

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$$S^{ff} = \prod_{j=1}^{J} \prod_{k=1}^{k_j} l_{jk}$$

Example : For two modes, three attributes of each mode and four levels per attribute

$$S^{ff} = \prod_{j=1}^{J} (4 \times 4 \times 4)$$

$$S^{ff} = \underbrace{(4 \times 4 \times 4)}_{\text{Mode 1}} \times \underbrace{(4 \times 4 \times 4)}_{\text{Mode 2}}$$

$$S^{ff} = 4096$$

The Core Idea



Attribute-level balance

Each level appears equally often



Zero correlation among design columns

Design columns are statistically independent

Independent estimation of main effects

For columns x_i and x_j, orthogonality means $x_i^T x_j = 0$ for $i \neq j$.

Orthogonal Design with Three Attributes

Orthogonal design with three attributes having two levels

s	A	B	C	AB	AC	BC
1	-1	-1	-1	1	1	1
2	-1	1	1	-1	-1	1
3	1	-1	1	-1	1	-1
4	1	1	-1	1	-1	-1
				$\Sigma = 0$	$\Sigma = 0$	$\Sigma = 0$

Correlation matrix

	A	B	C
A	1	0	0
B	0	1	0
C	0	0	1

Key Properties

1. **Each level appears equally often**
   Balanced representation of attribute levels across the design

2. **Columns are statistically independent**
   Zero correlation among design columns

3. **Easy to understand and communicate**
   Simple and intuitive structure for researchers and stakeholders

4. **Useful when priors are absent or weak**
   Provides a neutral, unbiased starting point

Strengths

- **Transparent structure** — Clear and systematic design construction
- **Balanced representation** — Each level appears equally often
- **Simple benchmark design** — Provides a solid baseline for comparison
- **Widely taught and widely understood** — Standard in textbooks and widely used in practice

Limitations

- **May generate dominated or unrealistic tasks** Some space design profiles may be implausible in practice
- **Orthogonality may not survive blocking or missing data** Balance and zero correlation can be disrupted
- **Less efficient than efficient designs** Typically requires more data for the same precision
- **Real behavior need not respect neat design balance** Human decision-making may violate design assumptions

Example

Modes ↓	Travel Time (TT) (mins)	Travel Cost (TC) (INR)	Waiting time (OVTT) (mins)
Public Transport (PT)	30,40,50	15,20,25	10,15,20
Private Vehicle (PV)	25,35,45	25,30,40	NA
Ride Hailing Services (RHS)	20,30,40	50,75,100	10,15,20
Auto Rickshaw (AT)	27,35,43	40,50,60	5,8,13

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Profiles	PT	PV	RHS	AT	Total
Full Factorial	$3 \times 3 \times 3 = 27$	$3 \times 3 = 9$	$3 \times 3 \times 3 = 27$	$3 \times 3 \times 3 = 27$	$27 \times 9 \times 27 \times 27 = 177147$
Orthogonal	Generated through IBM SPSS				27

Sample Choice card through Orthogonal Design

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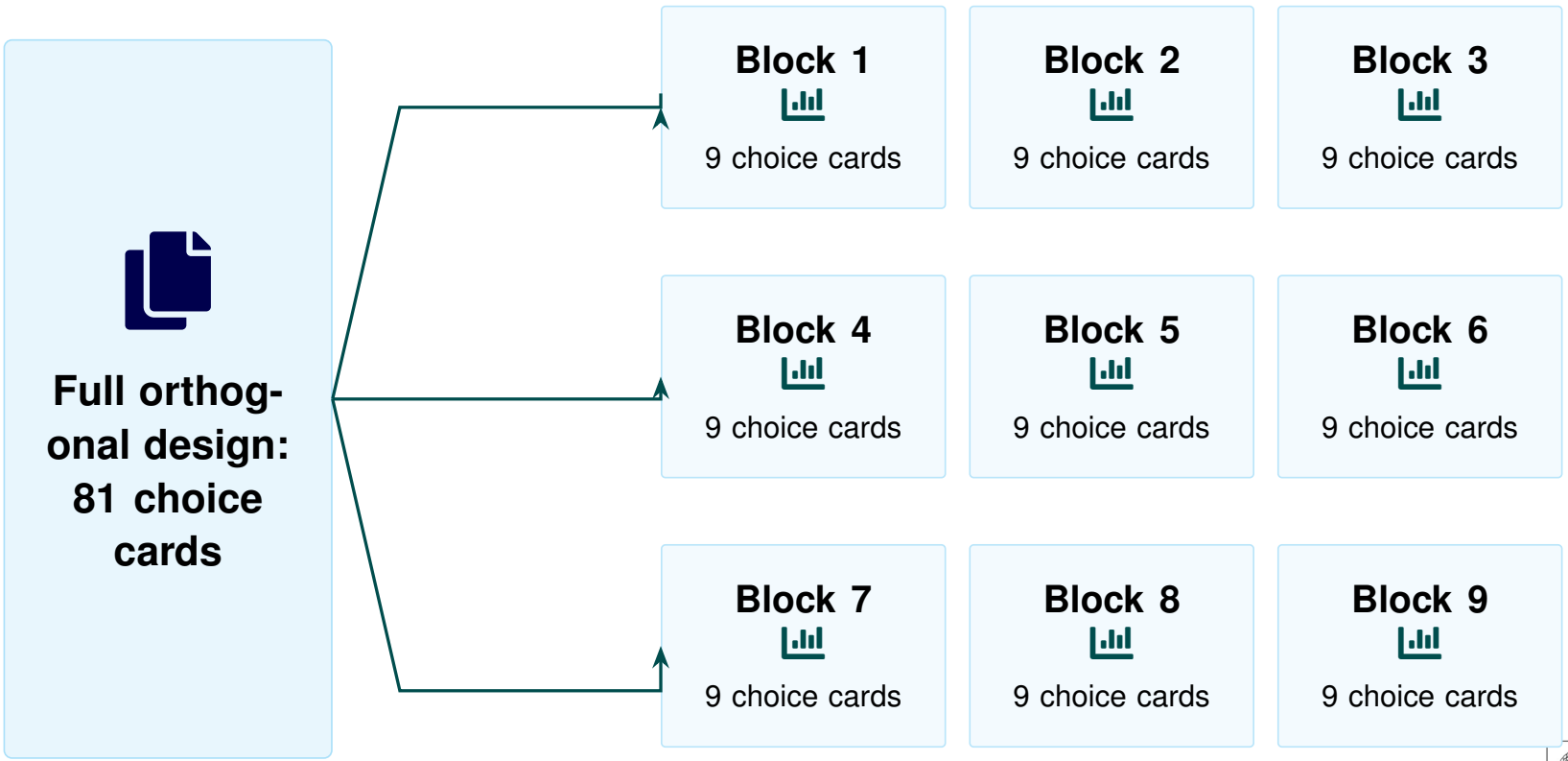
Sample Choice card through Orthogonal Design

PT_{TT}	PT_{TC}	PT_{OVTT}	PV_{TT}	PV_{TC}	RHS_{TT}	RHS_{TC}	RHS_{OVTT}	AT_{TT}	AT_{TC}	AT_{OVTT}
40	25	15	45	40	20	50	10	35	60	5

Full Orthogonal Design and Blocking

Requirement : An orthogonal design still may be too large to give all the choice situations to a single respondent.

Necessary condition : There should be attribute level balance in each blocks.



Properties of Orthogonal Blocks

What blocking means

1. Respondents answer only one block.
2. Individual blocks need not be orthogonal by themselves.
3. Orthogonality belongs to the complete design across all blocks.
4. Blocks should be assigned evenly across respondents.
5. Poor block administration can weaken the final realized orthogonality.

Attribute level balance

All available attributes

PT_{TT}, PT_{TC}, PT_{OVTT}, PV_{TT}, PV_{TC}, RHS_{TT}, RHS_{TC}, RHS_{OVTT}, AT_{TT}, AT_{TC}, AT_{OVTT}

Attribute level balance

All available attributes

PT_{TT}, PT_{TC}, PT_{OVTT}, PV_{TT}, PV_{TC}, RHS_{TT}, RHS_{TC}, RHS_{OVTT}, AT_{TT}, AT_{TC}, AT_{OVTT}

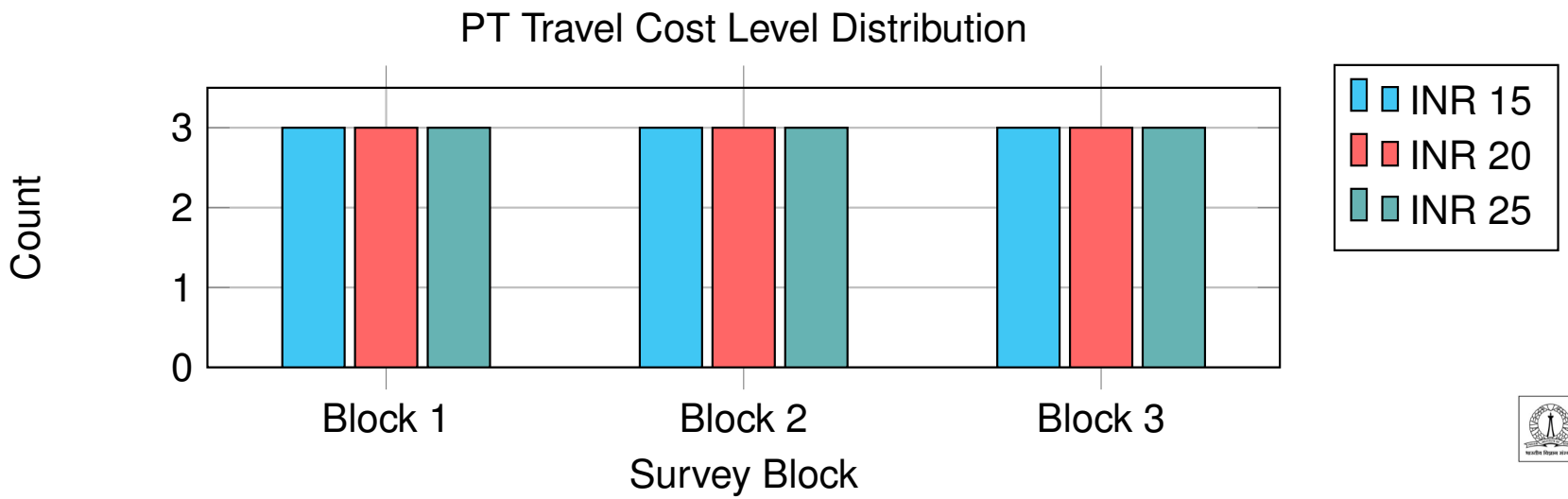
Attribute levels	Block 1 (9 cards)	Block 2 (9 cards)	Block 3 (9 cards)	Total (27 cards)
Level 1	3	3	3	9
Level 2	3	3	3	9
Level 3	3	3	3	9
Total	9	9	9	27

Attribute level balance

All available attributes

PT_{TT}, PT_{TC}, PT_{OVTT}, PV_{TT}, PV_{TC}, RHS_{TT}, RHS_{TC}, RHS_{OVTT}, AT_{TT}, AT_{TC}, AT_{OVTT}

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Level 1	3	3	3	9
Level 2	3	3	3	9
Level 3	3	3	3	9
Total	9	9	9	27



Available attributes and their levels

Attributes	PT	PV	RHS	AT
1. Travel Time (min)	0.75 TT 0.875 TT TT 1.125 TT 1.25 TT	0.75 TT 0.875 TT TT 1.125 TT 1.25 TT	0.75 TT 0.875 TT TT 1.125 TT 1.25 TT	0.75 TT 0.875 TT TT 1.125 TT 1.25 TT
Note: TT is taken from the revealed-preference (RP) trip.				
2. Travel Cost (Rs)	0.85 BFpt TD BFpt TD 1.15 BFpt TD	0.85 BFpv TD BFpv TD 1.15 BFpv TD	0.85 BFrhs TD BFrhs TD 1.15 BFrhs TD	0.85 BFat TD BFat TD 1.15 BFat TD
Note: BF = base fare for the respective mode; TD = travel distance from RP.				
3. Additional Surge Multiplier	1x	1x	0.9x, 1x, 1.25x, 1.5x, 1.75x, 2x	1x
Note: Total ride-hailing cost = Travel Cost + Additional Surge.				
4. Waiting Time (min)	5, 10, 15	0	5, 10, 15	2, 5, 8
5. Walking Distance (m)	100, 500, 1000	0	0	100, 250, 500
6. Number of Transfers	0, 1, 2	0	0	0

Attribute level Distribution

Different Design scenarios

Full Factorial : 221 million possibilities

Orthogonal Design : 81 Possibilities

Orthogonal Blocking : 3 blocks each having 27 choice cards (Maintaining attribute level balance)

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key Point in attribute level distribution : All the attributes are having same distribution across orthogonal blocks irrespective of the modes if it is available for that particular mode.

Attribute level Distribution

Different Design scenarios

Full Factorial : 221 million possibilities

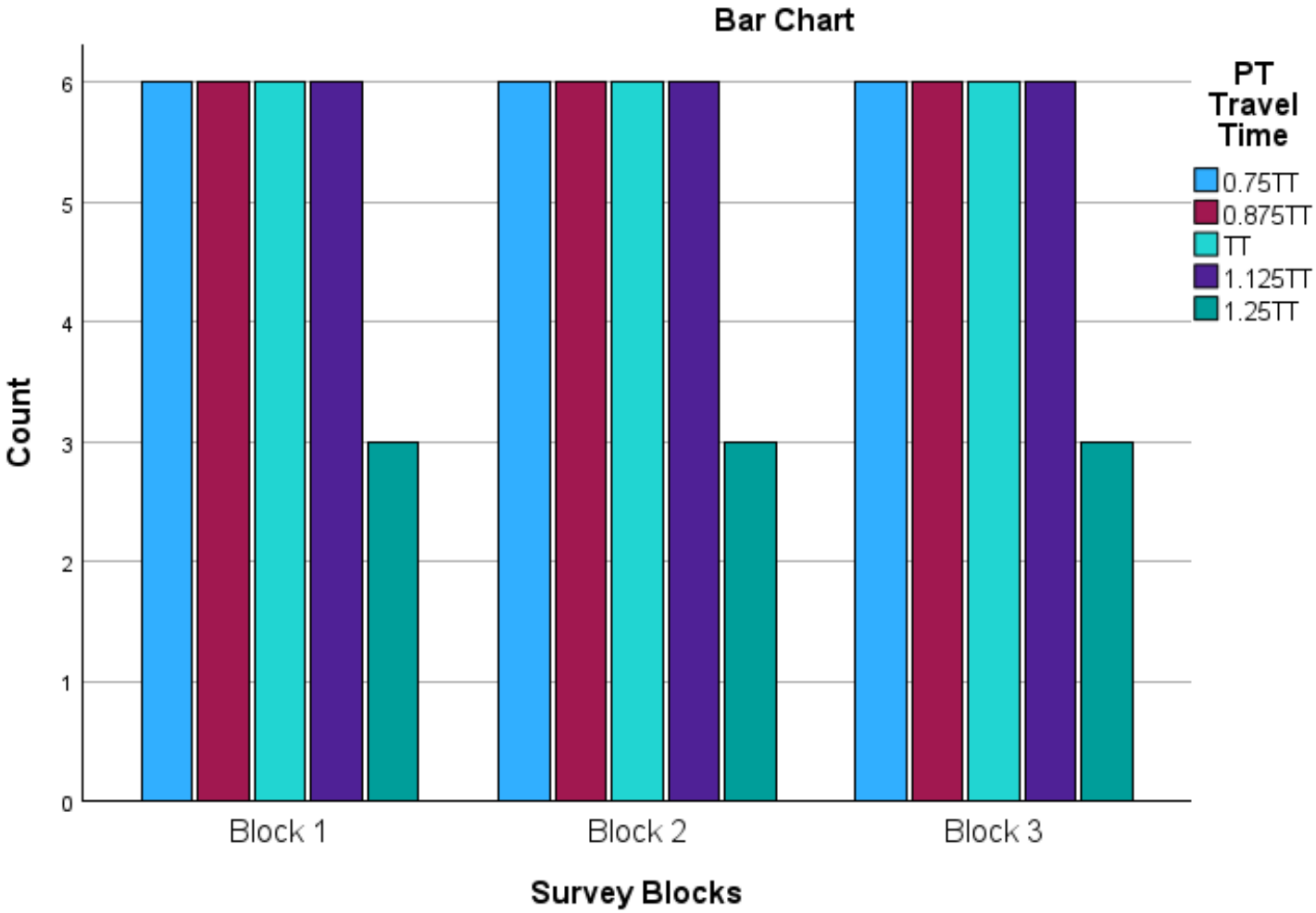
Orthogonal Design : 81 Possibilities

Orthogonal Blocking : 3 blocks each having 27 choice cards (Maintaining attribute level balance)

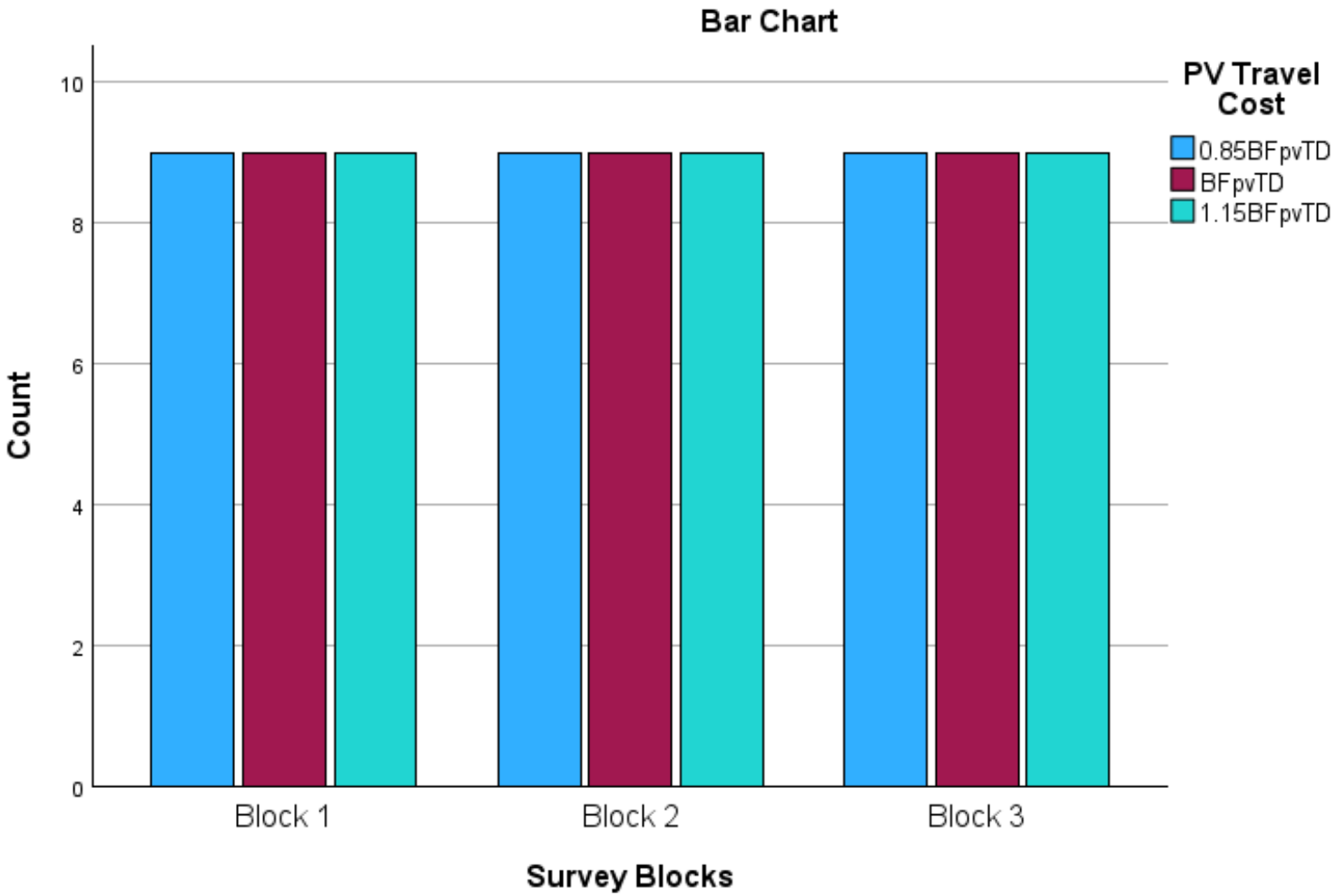
key Point in attribute level distribution : All the attributes are having same distribution across orthogonal blocks irrespective of the modes if it is available for that particular mode.

Attributes	Levels	Distribution across each blocks	Total
Travel Time (TT)	5	6,6,6,6,3	27
Travel Cost (TC)	3	9,9,9	27
Waiting Time (OVTT)	3	9,9,9	27
Walking Distance (WD)	3	9,9,9	27
Number of Transfers (NT)	3	9,9,9	27
RHS Surge	6	6,6,6,3,3,3	27

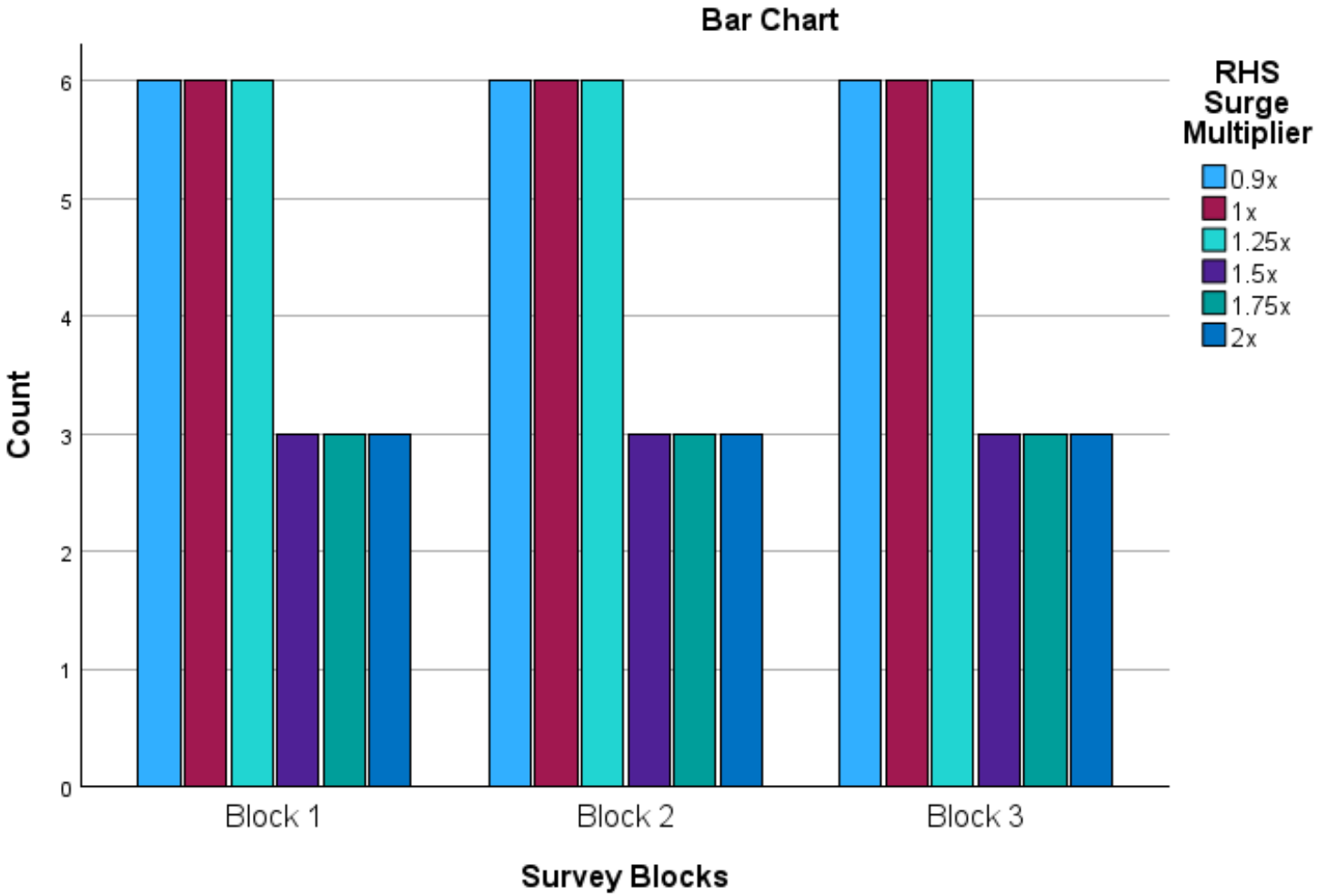
Travel Time


Bar Chart
PT Travel Time
0.75TT
0.875TT
TT
1.125TT
1.25TT
Count
0
1
2
3
4
5
6
Block 1
Block 2
Block 3
Survey Blocks

Travel Cost


Bar Chart
PV Travel Cost
0.85BFpvTD
BFpvTD
1.15BFpvTD
Count
10
8
6
4
2
0
Block 1
Block 2
Block 3
Survey Blocks

RHS Surge


Bar Chart
RHS Surge Multiplier
0.9x
1x
1.25x
1.5x
1.75x
2x
Count
0
1
2
3
4
5
6
Block 1
Block 2
Block 3
Survey Blocks





Thank You





ORTHOGONAL DESIGN: WHY LEVEL BALANCE ≠ ORTHOGONALITY?

Two conditions are required for an orthogonal design

1. ATTRIBUTE LEVEL BALANCE

Each level of every attribute appears equally often.

Attribute A (Cost)

Level (Code)	Frequency
Low (-1)	2
High (+1)	2

✔ Balanced

Attribute B (Time)

Level (Code)	Frequency
Short (-1)	2
Long (+1)	2

✔ Balanced

Balance looks at each attribute individually.

+

2. ORTHOGONALITY

The attributes are independent of each other.

$$A'B = 0$$

Knowing the level of Attribute A gives no information about Attribute B.

For coded attributes (-1, +1): Dot product of columns = 0

This is required for independently estimable parameters.

EXAMPLE 1: BALANCED BUT NOT ORTHOGONAL ✗

Card	Cost (A) (-1, +1)	Time (B) (-1, +1)
1	-1	-1
2	-1	-1
3	+1	+1
4	+1	+1

Level Balance

✔ Cost: 2 Low, 2 High
✔ Time: 2 Short, 2 Long

Dot Product (A'B)

(-1)(-1) + (-1)(-1) + (+1)(+1) + (+1)(+1) = 4 ≠ 0

Why Not Orthogonal?

Whenever Cost is Low, Time is Short.
Whenever Cost is High, Time is Long.

➜ Attributes move together (correlated).

NOT ORTHOGONAL

EXAMPLE 2: BALANCED AND ORTHOGONAL ✓

Card	Cost (A) (-1, +1)	Time (B) (-1, +1)
1	-1	-1
2	-1	+1
3	+1	-1
4	+1	+1

Level Balance

✔ Cost: 2 Low, 2 High
✔ Time: 2 Short, 2 Long

Dot Product (A'B)

(-1)(-1) + (-1)(+1) + (+1)(-1) + (+1)(+1) = 0

Why Orthogonal?

All combinations of Cost and Time appear equally.

➜ Knowing one attribute tells nothing about the other.

ORTHOGONAL

KEY DIFFERENCE

Level Balance	Checks each attribute separately (column totals).
Orthogonality	Checks relationship between attributes (dot product = 0).

TAKE-HOME MESSAGE

Balanced Attributes ≠ **Orthogonal Design** ⇒ **Orthogonal Design Requires Both!**

✔ We need both Level Balance and Orthogonality.
✔ Only then can parameters be estimated independently.